Utilizing Cashflow Underwriting for Fairer Credit Assessment

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Abstract

Traditional credit scores estimate an individual's likelihood of repaying loans but are limited to those with an existing credit history, leaving many groups underserved. To address this, we leverage natural language processing and large language models to categorize bank transaction memos and predict credit scores, incorporating income and cashflow data to better assess credit risk. We implement Logistic Regression, XGBoost, Random Forest, and LightGBM models to classify consumers as high-risk or low-risk for defaulting, and evaluate model performance using AUC, which measures the ability to accurately distinguish between these groups.

Code: https://github.com/elliekwang/DSC180A_B01_2

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1 Introduction

Traditionally, credit scores are determined solely by financial statement history, which focuses on assets, liabilities, and net worth. This approach can be unfair to many populations, such as low-income individuals, new graduates, contract workers, and immigrants, who may not have extensive credit records despite being financially responsible. Prism Data addresses this limitation by analyzing more than just traditional financial statements, incorporating income and transaction data to gain a fuller picture of financial behavior. Through its pioneering use of cash flow underwriting, Prism Data expands credit access to consumers who have limited or no traditional credit history, promoting a more inclusive and accurate assessment of creditworthiness.

1.1 Problem Statement

Given the banking transactions provided by Prism Data, our goal is to use feature engineering in transaction memos to create an accurate model that predicts credit risk at a rate of 90% or higher. Previous work has attempted to answer this question by modeling long sequences to parallelize training. RNNs, LSTMs, and GRUs capture sequential dependencies but require step-by-step processing, making training slow and unparallelizable. CNN-based models improved parallelism, but struggled to model long-range dependencies effectively. Subsequently, the transformer removed recurrence and convolution entirely, completely relying on self-attention and making computation parallelizable. Multi-head attention also allows the model to observe multiple perspectives simultaneously. Although attention applied to RNNs and CNNs was shown to improve accuracy, faster computational speed, and memory, the core compute limitations of sequential bottlenecks still existed. Self-attention was associated with a quadratic cost, which meant dealing with long sequences was computationally expensive. Large models also required massive amounts of compute and data, and interpretability is still limited since attentional weights do not reliably explain decisions.

Our project addresses the deficiency of high computational cost and interpretability by using DistilBERT, a transformer based model that inherits much of BERT's ability to model long-range dependencies through its self-attention mechanism. This allows us to avoid the sequential bottleneck problems of earlier RNN and CNN models, allowing us to fully parallelize longer sequences of text faster.

In a cash flow underwriting dataset, inflows show all the money coming into a consumer's account. The `consumer_id` identifies the person receiving the money, and the `account_id` shows which bank account it went into. The `memo` gives a short description showing where the money came from. The `amount` is how much was received, and the `posted_date` is when it was deposited. The `category` helps group the inflow by type, such as payroll, refund, or transfer. Together, these columns help lenders see how steady and reliable a person's income is (Figure 1).

Outflows, on the other hand, record all the money going out of the consumer's account. The

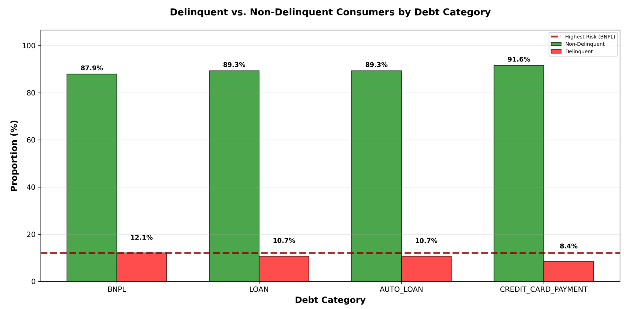

Figure 1: We used Pandas and Matplotlib to understand the delinquency rate among varying debt payment categories.

`consumer_id` and `account_id` again show who spent the money and from which account. The `memo` describes what the payment was for, such "as "RENT," "CREDIT CARD BILL", or "GROCERIES." The `amount` shows how much was spent, and the `posted_date` shows when the transaction occurred. The `category` classifies expenses such as housing, debt, or entertainment. Together, these columns reveal a person's spending habits and financial responsibilities, helping underwriters understand how much cash flow they have left after expenses.

The five categories we selected are `FOOD_AND_BEVERAGES`, `GENERAL_MERCHANDISE`, `GROCERIES`, `PETS` and `TRAVEL`. Within the `TRAVEL` category, the most common merchants are Uber, Uber Eats, and Lyft. For `PETS`, frequent merchants include PetSmart, CHEWY.COM, and Pet Supplies Plus. The `GROCERIES` category is dominated by Walmart, Kroger, and Target, while `GENERAL_MERCHANDISE` mainly includes Amazon, 7-Eleven, and Circle K. Lastly, in `FOOD_AND_BEVERAGES`, the most common merchants are McDonald's, Starbucks, and Chick-fil-A.

Beginning in January 2026, we were introduced to new datasets that provided information at the consumer, account, and transaction levels. Unlike the previous datasets we had dealt with from October to December 2025, these datasets were removed of the banking transaction memos for privacy protection reasons; we were directly provided what categories consumers made their purchases on. It is important to note that in a real setting, we would need to build a model to estimate the category from the banking transaction memos. Using the newly projected categories, we would build a model predicting whether a consumer would be delinquent before scaling their respective delinquency probabilities into an understandable credit risk value. The transaction data we dealt with arrived in the datasets as follows:

1. `consdf`: contains `prism_consumer_id`, `evaluation_date`, `credit_score`, and `DQ_TARGET`, where `DQ_TARGET` represents delinquent (1) or non-delinquent (0).
2. `acctdf`: contains `prism_consumer_id`, `prism_account_id`, `account_type` (e.g. savings, checking), `balance_date`, `balance`
3. `trxndf`: contains `prism_consumer_id`, `prism_transaction_id`, `category` (yields a numerical value), `amount`, `credit_or_debit`, `posted_date`

Since the `category` column in the `trxndf` dataset yields a numerical value, we determine the true name of the category (i.e., what it represents) using `cat_map`, which provides a mapping between the category ID and category (name).

2 Methods

To utilize cash flow underwriting to better estimate credit risk, we performed natural language processing on banking transaction memos. We had two datasets from Prism Data: a table of inflows containing 513,115 records, representing money that enters people's banking accounts, and an outflows table with 2,597,488 records, referring to expenses. Our data spanned from 2017 to 2023.

We preprocessed our data using regular expressions (Regex), carefully considering which words, phrases, and parts of speech to keep or remove. We filtered out irrelevant punctuation and long sequences of more than four characters in length. We also handled special cases, including but not limited to email addresses, parts of website URLs, street addresses, and possible variants for businesses with the same name.

For our banking transaction spending classification model, our baseline was a logistic regression model with Term Frequency–Inverse Document Frequency (TF-IDF) features. We also implemented FinBERT, a transformer pretrained on financial reports to capture domain-specific sentiment, and sought to improve this model by applying DistilBERT, another transformer pretrained through knowledge distillation to create a smaller and computationally faster model than BERT. Additionally, we experimented with a sentence encoder that generates fixed-length embeddings for each document. To further enhance the sentence encoder outputs, we trained an XGBoost classifier, combining the semantic representations with gradient boosting.

In Phase 2 of our project (Jan 2026 onwards), we were provided with the categories of each consumer's spending. Our goal is to develop a robust binary classification model to predict whether a consumer is high-risk or low-risk and to estimate the probability of loan repayment, which will later be scaled into a standardized and interpretable credit score. Before engineering features and building our model, we decided to restrict our data to the following criteria:

- Ignore consumers without an account.
- Ignore consumers without transactions.
- Consumers must have at least one credit and one debit transaction.
- Consumers should not have too short of observable history (i.e., less than 30 days).

We implemented Logistic Regression, XGBoost, LightGBM, and Random Forest using the top 50 most significant features, selected for their relevance to credit risk. Logistic Regression serves as a baseline due to its simplicity and interpretability, while gradient boosting models are included to capture non-linear relationships and complex interactions. Comparing these models allows us to evaluate the trade-off between interpretability and predictive performance, with the best-performing model identified using AUC.

To evaluate model performance, we use the Area Under the ROC Curve (AUC), which quantifies the model's ability to distinguish high-risk from low-risk consumers. AUC is particularly valuable because it assesses overall ranking performance across all possible decision thresholds, providing a threshold-independent measure of discrimination. We also considered precision, recall, and F1, but these metrics alone do not provide a complete picture of credit risk performance, as they are sensitive to the chosen classification threshold and may not fully capture the model's ability to rank consumers by risk.

3 Results

Our study evaluated four machine learning approaches for classifying consumers' delinquency: logistic regression, random forest, XGBoost, and LightGBM.

3.1 Baseline Model: Logistic Regression

For Logistic Regression, we configured the model with a maximum iteration of 1000, balanced class weights, and a random state of 42. After excluding accounts without transactions and accounts with less than 30 days of observable history and restricting the model to only accounts with at least one credit and one debit transaction, identifying the top 50 most predictive features, our refined model achieved an AUC of 0.757.

3.2 Bagging Model: Random Forest

After excluding accounts without transactions and accounts with less than 30 days of observable history and restricting the model to only accounts with at least one credit and one debit transaction, we identified the top 50 most predictive features for the Random Forest model. The refined model achieved a test AUC of 0.798, test accuracy of 0.916, test precision of 0.586, and test recall of 0.102.

3.3 Boosting Model 1: XGBoost

For XGBoost, we used 600 estimators with a shallow max depth of 4, slow learning rate of 0.01, and moderate subsampling. After excluding accounts without an account or transaction and accounts with too short of observable history (less than 30 days) and restricting the model to only accounts with at least 1 credit and 1 debit, we achieved a test AUC of 0.80, which is the highest AUC compared to any other model tested, except for LightGBM. After optimizing the threshold, the recall was about 0.376. The model performed well with precision, suggesting high confidence in flagging someone delinquent but missed correctly identifying most of the delinquents.

Since the train AUC was 0.934 compared to the test AUC of 0.801, it is reasonable to infer that some overfitting occurred despite implementing regularization parameters, including `reg_lambda=0.5`, `gamma=0`, and `min_child_weight=3`.

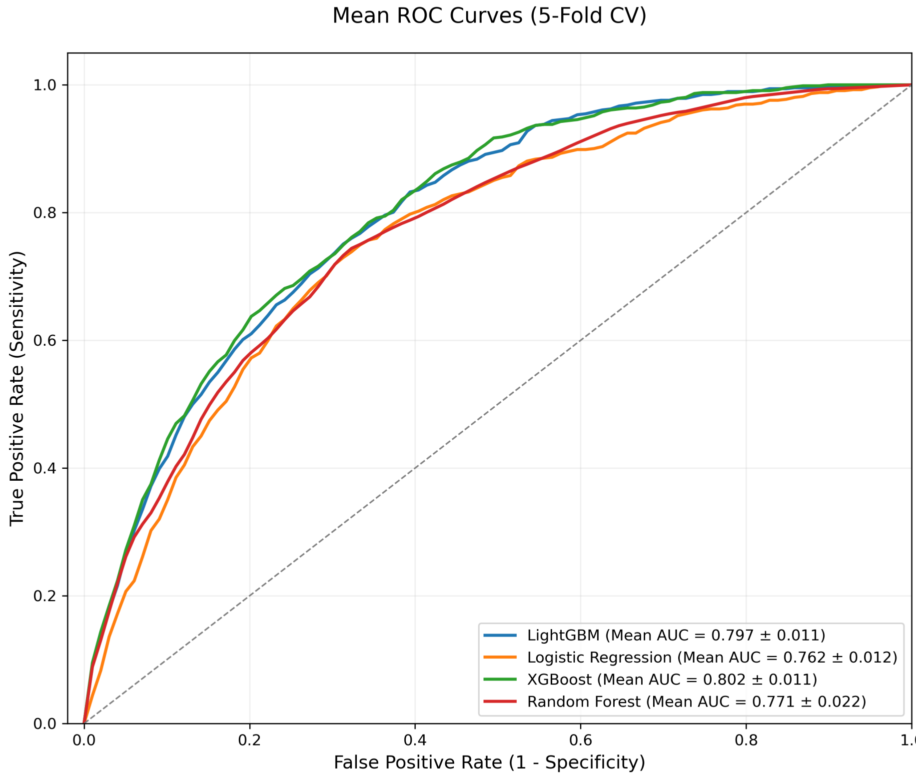

Figure 2: Mean ROC curves using 5-Fold CV for each of the models tested: LightGBM, Logistic Regression, XGBoost, and Random Forest. In terms of AUC, both LightGBM and XGBoost performed exceptionally well on the dataset.

3.4 Boosting Model 2: LightGBM

For LightGBM, we used 900 estimators with a shallow max depth of 3, slow learning rate 0.01, and conservative leaf count (15). To prevent overfitting on the minority class, we specified `min_child_samples=80`. We achieved a test AUC of 0.80, which had an improved precision-recall balance compared to XGBoost. Using the optimized threshold, we obtained a precision of 0.521 and a recall of 0.291. The Train AUC (0.903) and Test AUC (0.793) are also smaller in difference compared to XGBoost, suggesting that there is less overfitting. We also used `class_weight="balanced"` to handle the 8.6% delinquency rate in the training data. Using a 5-fold cross validation test, we also obtained a mean CV AUC of 0.798 (std 0.011), which demonstrates that the model generalizes consistently, rather than performs well in one occasion (Figure 2).

3.5 Summarized Results

The binned risk graph demonstrates how the predicted risk probabilities are distributed among consumers (Figure 3). For people with an average annual inflow below average (less

Table 1: Model Performance Comparison

Model	Accuracy	AUC	Precision	Recall	Inference Time (ms)
LightGBM	0.916	0.80	0.521	0.295	0.7261
XGBoost	0.886	0.80	0.357	0.398	0.77893
Random Forest	0.857	0.79	0.281	0.422	1.5259
Logistic Regression	0.776	0.76	0.209	0.572	15.70487

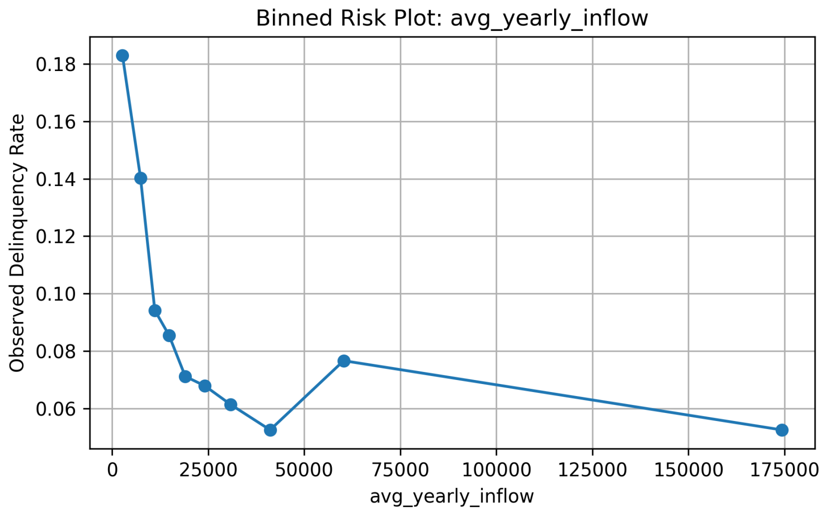

Figure 3: Negative trends in delinquency rates when binning consumers into 10 bins by average yearly inflow.

than $50,000 annually), there is an exponentially decreasing observed rate of delinquency, which stabilizes at around 6% once consumers reach a high annual yearly inflow above $75,000 per year (Figure 4).

4 Discussion

4.1 A General Comparison of the Models

Logistic Regression is simple, fast, and easily interpretable. It serves as a good baseline model, but assumes relationships between data points are linear, which means a logistic regression model struggles with capturing complex behavior and is liable to underperform on small datasets without resampling.

Random Forest is a tree-based ensemble method that handles more complex, high-dimensional, non-linear, and larger datasets compared to logistic regression. This model also tends to have better accuracy and is more robust to class imbalance but may be less interpretable. Using SHAP values can often help interpret random forest models. Random Forest also has a tendency to train slowly on large datasets and may overfit on noisy features more than single trees.

XGBoost is another highly optimized tree-based ensemble method that implements gradient boosting. It excels in classification, regression, and rank-based tasks, handles missing values natively, and handles imbalanced classes well. XGBoost models are highly tunable but may require more work than Random Forest models.

Like XGBoost, **LightGBM** is also a tree-based ensemble method that implements gradient boosting. It often matches or outperforms XGBoost in terms of training time, especially when dealing with larger datasets, and also handles imbalanced classes effectively. In other words, LightGBM is best suited for large datasets with high dimensionality where speed and performance are crucial.

4.2 Why LightGBM Excels

LightGBM has a high accuracy of 0.916, which means the LightGBM model correctly classified about 92% of all borrowers in the dataset. Although this metric may be volatile on imbalanced data, we also considered precision and recall to get a better understanding of our model's performance on correctly identifying delinquents. We obtained a precision of 0.521 and a recall of 0.295. In other words, out of everyone the model flagged as delinquent, about 53% were actually delinquent. A high precision is vital for minimizing false positives, which is essential to avoid denying credit to a financially smart consumer. Likewise, out of all actual delinquents, the model correctly captured about 30%. Using recall is vital to minimize false negatives to avoid lending to someone which a high risk of defaulting.

Both precision and recall require using a threshold. However, we don't want to assume the threshold credit score companies may be using, so we also evaluated AUC. AUC is indifferent to class distribution since it evaluates the model's ranking ability, or how well it separates the classes of delinquency across all possible thresholds, not just one. Our model's AUC of 0.80 suggests there is about an 80% chance a randomly chosen "good" (non-delinquent consumer) would have a higher credit risk score than a randomly chosen "bad" (delinquent consumer). In other words, the LightGBM model is pretty strong (80% chance) at effectively ranking a random delinquent as higher risk than a random non-delinquent.

With the LightGBM model, the strongest delinquency signals using the LightGBM model involve frequent internal transfers between accounts and income stability (Figure 3). Most consumers had a high feature value in low minimum monthly balance, which suggests most people had a low minimum monthly balance and had a lower delinquency risk. Conversely, higher minimum monthly balances were associated with a greater risk of delinquency (Figure 5).

4.3 Why Logistic Regression Underperforms

Logistic regression achieved a test AUC of 0.76, indicating fair-to-good discrimination ability. The model correctly ranks a delinquent case higher than a non-delinquent case approx-

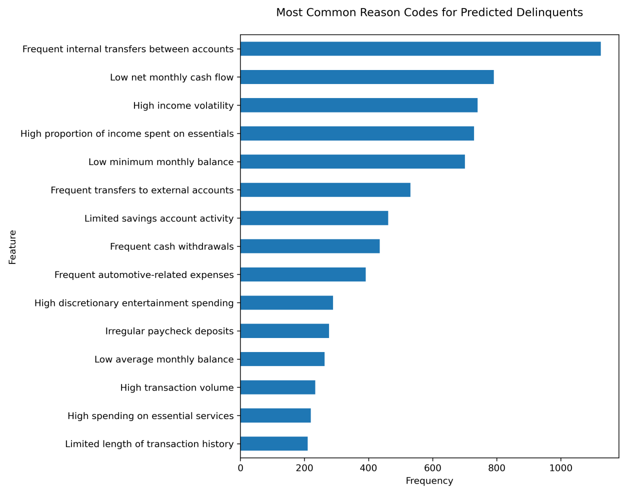

Figure 4: Bar graph of the most common reason codes for predicted delinquents.

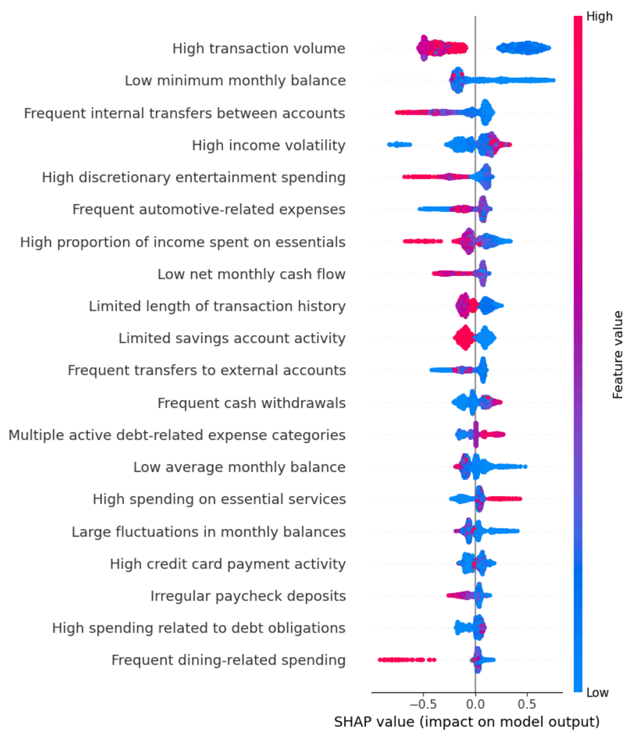

Figure 5: Negative values suggest a lower risk of delinquency and positive values suggest a higher risk of delinquency; blue dots indicate lower feature values and red dots indicate higher feature values. Monthly balance, internal transfer between accounts, and entertainment spending were among some of the key signifiers that impact predicted risk of delinquency.

imately 76% of the time. While reasonable for financial classification tasks, this represents the lowest performance among our tested models, suggesting room for improvement.The underperformance stems from logistic regression's fundamental assumption of linear relationships between features and the log-odds of delinquency. Financial data, however, exhibits complex non-linear patterns and feature interactions that logistic regression cannot capture. For example, the relationship between income volatility and delinquency may not be purely linear. Similarly, interactions between features such as low income combined with high debt payments may have non-linear effects on delinquency risk that logistic regression overlooks.

4.4 The Limitations of Random Forest

The high accuracy of 0.857 is misleading due to class imbalance, as the model achieves this by predominantly predicting the majority class. The precision of 0.281 indicates that when the model predicts an account as delinquent, it is correct 28% of the time. The recall of 0.422 reveals that the model identifies 42% of actual delinquent accounts, which, while moderate, still means the majority of at-risk cases go undetected. Compared to the chosen model's recall of 0.295, Random Forest captures more delinquent accounts at the cost of lower precision, reflecting a trade-off between sensitivity and specificity. Random Forest's AUC of 0.79 outperformed Logistic Regression 0.76 due to its ability to capture non-linear relationships, but was surpassed by LightGBM and XGBoost, which use stronger regularization and more effective error correction for improved generalization.

4.5 A Comparison of XGBoost and LightGBM

Both models perform similarly and are very comparable, depending on the inputted parameters, chosen random state, threshold selected, etc. Since we cannot make an inference on the threshold credit score companies would be using, our primary metric of evaluation is AUC. Both XGBoost and LightGBM achieved an AUC of 0.8. This means both models equally have an 80% chance of correctly stating a randomly chosen non-delinquent consumer will have a higher credit score than a randomly chosen delinquent consumer.

Since the inference time for LightGBM was 0.7261 ms compared to 0.77893 ms for XGBoost, LightGBM offered a slight improvement in inference time compared to XGBoost. Considering the performance between XGBoost and LightGBM were quite similar, we ultimately decided to select LightGBM for its faster speed.

4.6 Limitations

The model relies on transaction data, often captured as screenshots, which offer only a partial representation of a consumer's financial history. Missing, incomplete, or ambiguous memos, high-volume repetitive transactions, seasonal shifts such as tax season or holidays, and unseen transaction types can reduce accuracy. There is a risk of misclassifying consumers as delinquent or nondelinquent. False positives may unfairly restrict access to credit, while false negatives may expose lenders to financial loss. Performance also depends on the populations and transaction patterns represented in the training dataset and may not generalize well to unseen data.

4.7 Implications

A report by the Consumer Financial Protection Bureau estimates that about 26 million Americans are "credit invisible." PrismData enhances traditional credit scoring by applying natural language processing to banking transaction memo data to extract insights into consumer spending behavior and improve credit risk estimates. This approach can expand credit access for underserved populations, including immigrants, young adults, recent graduates, and gig workers, while providing lenders with more accurate assessments of borrower risk. By incorporating alternative data, the model has the potential to promote financial inclusion and support fairer lending practices.

5 Conclusion

Our work demonstrates how modern machine learning models can improve credit risk assessment for underserved populations. Using cash-flow underwriting—analyzing transaction patterns rather than traditional credit history—we developed classification models to identify high-risk and low-risk consumers. After preprocessing transaction data with Regex and DistilBERT for transaction classification, we tested four credit risk models: Logistic Regression, Random Forest, XGBoost, and LightGBM. LightGBM emerged as the strongest performer. Our findings validate that alternative data sources combined with modeling approaches can provide a more comprehensive assessment of creditworthiness than traditional scoring frameworks. These results support the potential of cash-flow underwriting to broaden credit access for individuals with limited credit history while providing financial institutions with more accurate risk evaluation. By incorporating more feature considerations, we demonstrate a scalable path toward fairer, more inclusive credit decisioning that benefits underserved populations.